

Larry Miguel R. Cueva

larrymiguelcueva@gmail.com | (+63) 970 745 1021 | <https://www.linkedin.com/in/michaelcueva> | <https://github.com/08Aristodemus24>

SUMMARY

A results-driven Data Engineer at ACEN with ownership of two end-to-end pipelines built from the ground up: the ACES Cost Methodology ELT pipeline and the IT Group Monthly Report pipeline. Currently spearheading a long-term ACES redesign with SharePoint Lists, MS Graph API, and Snowflake Cortex Analyst. Proven track record of automating processes to eliminate costs, monitor KPIs that drive improve work output, and enabling data-driven decision making for finance and IT stakeholders.

EXPERIENCE

ACEN Corporation <i>Data & Analytics Engineer</i> ACES Cost Methodology Pipeline	Dec 2025 – Present
- Architected and built a novel ELT pipeline, ingesting quarterly cost allocation data from Excel workbooks spanning 3 fiscal years (2024–2026) across 16+ workstreams and 4 cost categories (Committed, Optimized, Shared, Transactional) into Snowflake via Azure Blob Storage, replacing a fully manual Excel-driven process and automating cost allocation reporting for the FP&A team. - Developed a complete 3-layer dbt transformation pipeline (staging → intermediate → marts) for ACES cost data, engineering entity × workstream × period fact tables, SCD Type 2 snapshot dimensions to accurately track quarterly workstream cost category migrations and mid-year entity transfers — surfaced in a Power BI dashboard with YOY cost and volume analysis for FP&A and CFO-level stakeholders. - Spearheading the long-term ACES cost pipeline redesign — replacing fragile Excel column-based extraction with distributed SharePoint Lists → MS Graph API → Azure Blob Storage → Snowflake architecture, eliminating analyst bottlenecks from manual data pivoting, enabling zero-code-change handling of quarterly workstream cost category shifts, and delivering Snowflake Intelligence Agents as the self-service interface for FP&A users. IT Group Monthly Report Pipeline	
- Architected and implemented a zero-cost IT group data pipeline by replacing a paid Fivetran connector with a custom REST API integration (SDP) using enterprise-tier OAuth, reducing data ingestion connector costs to \$0 and eliminating recurring SaaS spend. - Automated IT support tickets, Microsoft user information, and DBT models meta data ingestion, into Snowflake via SDP Cloud, Microsoft Graph, and DBT Discovery APIs, enabling real-time analytics across 5+ IT group teams and processing thousands of ticket records daily with zero manual intervention. - Developed a machine learning time-series forecasting model scheduled to predict IT support ticket volumes 1–6 months ahead every month, enabling proactive resource planning and reducing response lead time by equipping IT management with forward-looking demand signals. - Engineered end-to-end Power BI dashboards tracking KPIs — total ticket requests, avg. support duration, SLA compliance rate by team, and YOY ticket trends — consolidating visibility across 5+ IT groups and surfacing actionable insights for IT leadership. Project Apollo — SAP ERP & ELT Contributions	
- Contributed to the Project Apollo SAP ERP ingestion pipeline by engineering a pandas preprocessing layer for raw SAP .txt files, standardizing 14 source table schemas (ACDOCA, BKPF, CSKS, SKA1, BSIK, etc.) and creating Snowflake file formats, external stages, raw tables, and Snowpipe-based ingestion from Azure Blob Storage. - Implemented MD5-hashed composite primary key MERGE logic across 14 SAP source tables to ensure idempotent, deduplicated incremental loads, and authored corresponding dbt staging model definitions in <code>_sap__sources.yaml</code> for the SAP data layer.	
Virtuals Protocol <i>Data Engineer, Intern</i>	Dec 2024 – Jan 2025
- Cleaned and processed more than 500k rows of data for various retrieval augmented generated (RAG) AI agents. - Developed and wrote scripts automating data ingestion processes of RAG AI agents and pulling raw datasets uploaded by users diverting main workflow to data transformation.	

PROJECTS

eda-denoiser-stress-detector | *React.js, D3.js, Flask, Scikit-Learn, Tensorflow, Docker*

- Enhanced the accuracy and reliability of bio-signal denoising and stress detection by developing a novel hybrid LSTM-SVM deep learning model, addressing critical challenges in bio-signal data analysis. Link to research: <https://aristodemus8-eda-denoiser-stress-detector.hf.space/>
- Engineered and deployed a full-stack web application demonstrating the utility and potential of the validated LSTM-SVM model in real world health monitoring applications.
- Validated model performance of 90% AUC & 78% accuracy in biosignal denoising, providing a robust foundation and methodology for future bio-signal research and potential diagnostic tools.

signal-gender-predictor | *SQL, DuckDB, Librosa, Azure Data Lake, Azure Data Factory, Airflow, Terraform*

- Developed an end-to-end MLOps pipeline for a gender prediction model API based on audio signals, reducing cloud operational costs by over 70%, leveraging cloud only for compute during extraction and storage. Link to API: <https://aristodemus8-signal-gender-predictor.hf.space/>
- Automated the ingestion, transformation, and feature engineering of large-scale audio datasets, generating high-quality features for model training and serving.
- Architected and implemented a scalable multi-stage data pipeline to efficiently process 3.7 billion rows of audio signals, generating high-impact features for voice-based gender prediction, providing a framework for real-time audio analytics and a foundation for voice-based AI applications.

chronic-disease-analyses | *SQL, Power BI, Apache Spark, Airflow, Selenium, S3, DuckDB, Docker, Terraform*

- Processed and transformed 20 years of comprehensive US public health data (from 2001-2021) using Spark, consolidating disparate datasets to quantify chronic disease cases and population figures. Link to project: <https://chronic-disease-analyses.vercel.app/>
- Conducted analysis of chronic disease data to identify most prevalent disease, allowing for potential in more targeted interventions and improving cost efficiency for less prevalent diseases

TECHNICAL SKILLS

Core Competencies: Data Engineering | ELT/ETL | Data Warehousing | Data Modelling | Data Orchestration | Cloud Infrastructure | Machine Learning | APIs | CI/CD | Data Analysis

Languages & Tools: Python | SQL | Snowflake | dbt | Power BI | Azure Cloud Services | Spark | Cortex Code | Git

EDUCATION

Polytechnic University of the Philippines
Bachelor of Science in Computer Science

Aug 2019 – Mar 2025

- 1.9 GPA